# Desk Research

**Project working title:** Rail Pulse - Understanding Passenger Reactions to Railway Disruptions

**Country:** The Netherlands

**Infrastructure focus:** Dutch passenger railways, particularly major disruptions related to railway infrastructure

**Proposed tool:** A live dashboard that uses a pretrained Hugging Face transformer to classify the sentiment of public posts about railway disruptions.

## 1. Problem statement

Background

Reliable railway infrastructure is important for a country's transport system. When railway infrastructure fails, trains may be delayed or cancelled, and passengers can experience difficulties reaching work, university, appointments or other destinations.

This is a documented problem in the Netherlands. According to ProRail's 2025 annual report, the country experienced 586 high-impact infrastructure disruptions in 2025, compared with 507 in 2024. This exceeded ProRail's benchmark of 520 disruptions.

ProRail identified the main causes as:

- 42% technical problems, including issues with switches, train detection, track condition and power supply.
- 44% third-party incidents, including people, traffic and animals on the tracks.
- 10% process-related causes, including overruns and problems following engineering works.
- 4% weather-related causes, including storms, lightning and extreme temperatures.

ProRail explains that it is working to reduce disruptions through more targeted maintenance and other measures.

Why is this a problem?

1. Disrupted passenger journeys

Train disruptions can interrupt journeys, increase travel times and cause passengers to miss connecting trains or arrive late at their destinations.

2. Passenger frustration and uncertainty

Passengers may become frustrated when their journeys are interrupted, particularly when they receive insufficient information about the cause, expected duration or alternatives.

3. The importance of communication

Research into unplanned passenger rail disruptions has found that passengers need reliable, up-to-date information that communicates the situation clearly and acknowledges the impact of delays on their journeys.

The problem this project addresses

Railway disruptions are already monitored through operational systems. However, information about an incident and information about how passengers are reacting to it are two different things.

For example, an operational dashboard may indicate that a technical failure has disrupted train services. Passengers' public posts may reveal concerns about missed appointments, uncertainty, the quality of travel information or the duration of the disruption.

The project focuses on these expressed passenger reactions.

Problem statement

Railway infrastructure disruptions affect the reliability of passenger journeys in the Netherlands. Although railway operators and infrastructure managers monitor operational incidents and provide travel information, understanding passengers' reactions during disruptions is also important. This project investigates how live sentiment analysis of public online posts can help a railway communications or service-quality team understand passenger concerns during major railway disruptions.

Research question

How does public sentiment expressed in online posts change during major railway disruptions in the Netherlands, and what passenger concerns appear during these events?

The project will focus on the sentiment expressed in public posts relating to railway disruptions. It will not attempt to diagnose infrastructure faults or assume that every train delay is caused by infrastructure.

## 2. Sustainable Development Goal (SDG)

Primary SDG: SDG 9 - Industry, Innovation and Infrastructure

The United Nations describes SDG 9 as building resilient infrastructure, promoting inclusive and sustainable industrialisation, and fostering innovation.

The most relevant target is Target 9.1:

Develop quality, reliable, sustainable and resilient infrastructure, including regional and transborder infrastructure, to support economic development and human well-being, with a focus on affordable and equitable access for all.

How does this project relate to Target 9.1?

Railways are part of a country's transport infrastructure. When infrastructure failures disrupt train services, they affect the reliability of that system and the journeys that depend on it.

The project relates to this target in three ways:

- Infrastructure reliability: The underlying problem is disruption to railway services, including disruptions resulting from technical infrastructure failures.
- Human well-being: Railway disruptions can negatively affect passengers' daily activities and travel experiences.
- Innovation: The project explores a technology-based method for analysing public feedback alongside existing operational information.

The dashboard could help a railway team identify changes in passengers' expressed concerns and investigate significant increases in negative reactions.

What the project will not claim

The system does not repair infrastructure, prevent failures or directly measure whether the Netherlands has achieved Target 9.1. Sentiment is an additional source of information about passengers' experiences, not an official SDG indicator.

SDG justification: The project supports the broader objective of reliable infrastructure by exploring how passenger feedback can contribute to understanding the public impact of railway disruptions.

## 3. User group

**Primary user:** Railway disruption communications or service-quality analyst

The proposed primary user is an employee involved in passenger communication, disruption management or service-quality analysis at a Dutch railway organisation, such as NS or ProRail.

This is a proposed user group for the project; it is not based on an interview or a confirmed request from either organisation.

What does this user need?

During a major disruption, the user may need to understand not only the operational situation but also the concerns being expressed by passengers.

Possible needs include:

- Identifying sudden increases in negative public reactions during a disruption.
- Understanding the reasons passengers express frustration.
- Reviewing example posts to distinguish complaints about delays, cancellations, uncertainty and communication.
- Comparing changes in public sentiment with known operational incidents.
- Recognising when a particular disruption appears to generate recurring passenger concerns.

What can the user do with the information?

For example, a dashboard could show an increase in negative sentiment while a major disruption is affecting train services.

The analyst could inspect the posts to understand the context and check the information against official disruption updates. If passengers repeatedly express confusion about alternatives or restoration times, the analyst could assess whether clearer updates are needed.

This would make the dashboard a decision-support tool. It would not independently determine the cause of an incident or decide how railway engineers should repair infrastructure.

**Secondary user group:** Passengers

Passengers are the people directly affected by railway disruptions. Their publicly expressed experiences provide the information being analysed. They are not necessarily direct users of the dashboard.

## 4. Existing solutions and approaches

### A. ProRail's performance dashboard

ProRail publishes information about railway performance, including infrastructure disruptions, punctuality, average disruption-recovery time and events that have caused significant passenger inconvenience.

This helps users understand operational performance and the history of major incidents.

Limitation in relation to this project: Operational information describes incidents and performance. It does not, by itself, classify the sentiment expressed in public passenger posts.

### B. NS travel information and APIs

NS provides travel information to passengers and offers APIs for developers. Its API portal includes data such as timetables, engineering works and station information.

These services support travel planning and access to operational information. Depending on availability and access requirements, official information could also provide context for interpreting social-media trends.

### C. Social media during railway disruptions

Research by Pender and colleagues examined how railway agencies use social media during unplanned passenger disruptions. The study highlights the need for reliable, current and clear information that recognises the impact of delays.

This provides a direct link between railway disruptions, passenger communication and public online discussions.

### D. Research into passenger satisfaction and rail delays

A 2021 study, *Modelling the impact of rail delays on passenger satisfaction*, combined passenger-satisfaction responses with operational data to investigate how delays affect satisfaction.

This supports the relevance of studying the passenger experience alongside operational railway performance.

### E. Proposed contribution of Rail Pulse

The project will combine a live stream of public posts with automated sentiment classification and a dashboard showing changes over time.

Its proposed contribution is to make passenger reactions easier to observe alongside operational disruption information.

This is not the same as claiming that no railway organisation already monitors social media or passenger complaints. The precise added value of our prototype will need to be demonstrated through its design and evaluation.

## 5. Ethical considerations

### A. Representativeness

Not every passenger posts online. People who post during disruptions may be more frustrated, more active on social media or more likely to share unusual experiences.

Therefore, negative sentiment in the collected posts cannot be treated as the opinion of all Dutch train passengers.

Safeguard: Show the number of posts analysed and describe the results as sentiment among collected public posts, not passenger-wide satisfaction.

### B. Model accuracy and language

Sentiment models can misunderstand Dutch slang, sarcasm, humour, negation and mixed opinions. A post may also express frustration without clearly stating whether the cause was infrastructure, service communication or another issue.

Safeguard: Test the model using human-labelled posts, record incorrect predictions and show representative examples. Posts should be filtered for relevance to railway disruptions before conclusions are drawn.

### C. Distinguishing sentiment from the cause of a disruption

A negative post about a delayed train does not prove that a signal, switch or other infrastructure component failed. Disruptions can have several causes.

Safeguard: Compare social-media patterns with reliable operational information where available. Do not use sentiment alone to assign responsibility or identify engineering faults.

### D. Privacy

Even publicly available posts may contain usernames, personal details or information about individual journeys.

Safeguard: Collect only what is necessary, avoid profiling individual passengers, minimise stored identifying information and follow the relevant platform rules and data-protection requirements.

**E. False alarms and misinterpretation**

A sudden change in sentiment may result from a small sample, a viral post or an unrelated event.

Safeguard: Display post counts, inspect the underlying messages and require human review before taking action. Do not treat a sentiment spike as proof that an infrastructure problem has worsened.

**F. Fairness and responsible decisions**

The system should support understanding and communication rather than automatically blame staff, rank individual passengers or determine maintenance priorities.

Safeguard: Use sentiment as one source of contextual information, alongside operational evidence and human judgement.

## 6. Feasibility for the later project

Proposed live data source: **Bluesky Jetstream**

Bluesky provides a service called Jetstream that streams public network events, including newly published posts, over a WebSocket connection. The data arrive in a structured format that a Python application can process.

For the prototype, the intended approach is:

1. Connect to the live stream of public posts.
2. Filter posts locally using relevant Dutch and English terms about railway disruptions, such as *treinstoring*, *treinuitval*, *treinvertraging*, *spoorstoring*, *wisselstoring* and *bovenleiding*.
3. Remove irrelevant matches, duplicates and posts that do not concern railway disruptions.
4. Run a pretrained Hugging Face transformer locally to classify sentiment.
5. Store the relevant text, timestamp and model prediction, with identifying data minimised.
6. Display sentiment changes over time and allow the user to inspect the example posts behind a peak or dip.

Jetstream filters the stream by record type and account, rather than directly filtering arbitrary words in post text. The application must therefore filter the text locally.

Will there be enough live posts?

The live source exists and its documented functionality fits the technical idea. However, the number of Dutch-language posts specifically about railway disruptions has not yet been measured.

The project should run a short collection test before the main data-collection period. If the number of relevant posts is too low, the search should cover a broader range of Dutch railway disruptions rather than only technical infrastructure failures.

This is the remaining data-volume uncertainty. It does not undermine the problem research, but it needs to be resolved before promising a specific number of relevant posts.

**Sentiment model**

The hackathon requires a pretrained Hugging Face transformer to run locally. A multilingual sentiment model is a possible starting point, but its accuracy on Dutch railway-related posts must be tested rather than assumed.

The evaluation should include a manually labelled sample, comparison against another baseline and examples of model failures, in line with the assignment requirements.

## 7. Conclusion

Railway infrastructure disruptions are a documented issue in the Netherlands. ProRail's 2025 annual report recorded 586 high-impact infrastructure disruptions, exceeding its benchmark. The problem is directly relevant to the broader aim of SDG 9, Target 9.1, which concerns reliable and resilient infrastructure.

The proposed project focuses on a related but distinct question: how passengers publicly react when railway disruptions affect their journeys.

A live public-post stream and a locally run sentiment model provide a plausible way to build the prototype. The dashboard could help a proposed railway communications or service-quality analyst recognise changes in public concern, examine the underlying posts and assess whether follow-up communication is needed.

The tool will not diagnose infrastructure failures or replace official incident-management systems. Its purpose is to help interpret the passenger impact of disruptions using public sentiment, while keeping uncertainty and model limitations visible.

## Sources:


ProRail Annual Report 2025 - Reliable railways.

Pender, B., Currie, G., Delbosc, A. and Shiwakoti, N. (2014). International Study of Current and Potential Social Media Applications in Unplanned Passenger Rail Disruptions.

United Nations - Goal 9.

ProRail Performance Dashboard.

NS API Developer Portal.

Pender et al. (2014).

Modelling the impact of rail delays on passenger satisfaction.

Official technical source: Bluesky Jetstream documentation.